Final Project Report

Direct Stiffness Method Program for 3D Structural Analysis

Figure 1: Bridge photograph for the final complex-structure study

Abstract

This project implements a Direct Stiffness Method program for the analysis of three-dimensional truss, frame, and mixed truss-frame structural systems. The code reads a structural model from a JSON input file, assembles global stiffness and equivalent load terms, solves for nodal displacements and support reactions, and exports tabulated postprocessing results to Excel. The repository includes separate solvers for truss, frame, and hybrid models, together with a notebook-based workflow that documents validation studies and a final bridge application.

The report is organized into three parts: code structure and input format, technical validation, and application to a complicated bridge model. First, it explains how the program is structured, how the JSON input is organized, and how the code can be executed and adapted for other structural models. Second, it presents validation studies for a 3D truss model and a 3D frame model. Third, it applies the code to a larger mixed system: a three-span arch-truss bridge model in which most members are idealized as truss elements while the arch ribs and the longitudinal members are modeled as frame elements.

Overall, the project demonstrates a consistent DSM workflow from model definition through solution and postprocessing. The validation studies and the final bridge application show that the implementation can be used to analyze structural response, organize results, and support interpretation of load paths and critical response patterns.

1 User Guide for the Code

1.1 Software environment and required packages

The project is configured through an environment file and is intended to be run in a Jupyter or VS Code workflow. The environment file specifies Python 3.12.12 together with NumPy, SciPy, Matplotlib, Plotly, Pandas, ipykernel, JupyterLab, and openpyxl. These packages are sufficient for numerical assembly and solution, notebook execution, visualization, tabular postprocessing, and Excel export.

Table A1. Software environment specified in environment.yml

Package	Version	Purpose
Python	3.12.12	Interpreter version used for the project
numpy	2.4.0	Array and matrix operations
scipy	1.16.3	Scientific utilities and numerical support
matplotlib	3.10.8	Static plotting support
plotly	6.5.2	Interactive structural visualization
pandas	2.3.3	Tabular result processing
ipykernel	7.1.0	Notebook kernel support
jupyterlab	4.5.1	Notebook environment
openpyxl	3.1.5	Excel export for output tables

1.2 Repository organization

The project files are organized into modules for input, element routines, assembly, solution, and postprocessing. The main notebook serves as the first level driver, while the helper modules separate the FEM workflow into element routines, assembly, partitioning, solvers, interfaces, preprocessing, and postprocessing. Input JSON files are grouped by validation and final-study load case, and exported Excel tables are stored in the outputs directory.

Table A2. Main project folders and files

Location	Role in workflow
main.ipynb	Notebook driver used to run the validation studies and final bridge cases
helpers/interfaces.py	Reads JSON blocks such as nodes, elements, constraints, and loads
helpers/element.py	Element-level geometry, stiffness, transformations, and DOF maps
helpers/assembly.py	Global stiffness and equivalent-load assembly
helpers/partition.py	Partitions free and restrained DOFs
helpers/preprocess.py	Equivalent nodal loads, moment release, and fictitious DOF handling
helpers/solver.py	Contains truss_solver, frame_solver, and hybrid_solver
helpers/postprocess.py	Builds summary tables, plots, and Excel exports
inputs/	JSON files for validation and final bridge load cases
outputs/	Excel output files generated by the notebook
examples/	DXF geometry files for the three models

1.3 How to run the program

The analysis is reproduced by running **main.ipynb** sequentially. The notebook imports helpers.solver as solver and helpers.postprocess as postprocess, then solves each input file through one of the three solver entry points. Truss validation files are solved with solver.truss_solver, frame validation files are solved with solver.frame_solver, and the final bridge load cases are solved with solver.hybrid_solver.

Each analysis case follows the same execution pattern: define a filepath string, run the corresponding solver, then pass the solver result to the matching postprocess function: postprocess.truss_output for truss models, postprocess.frame_output for frame models, or postprocess.hybrid_output for mixed truss-frame models. When table_output='xlsx' is selected, the code writes Excel tables to the outputs folder while also retaining the returned pandas DataFrames and figure objects for notebook display.

Table A3. Standard execution pattern in the notebook

Model type	Solver call	Postprocess call
3D truss validation	solver.truss_solver(filepath)	postprocess.truss_output(filepath, result, table_output='xlsx')
3D frame validation	solver.frame_solver(filepath)	postprocess.frame_output(filepath, result, table_output='xlsx')

Model type	Solver call	Postprocess call
Mixed truss-frame bridge	solver.hybrid_solver(filepath)	postprocess.hybrid_output(filepath, result, table_output='xlsx')

1.4 JSON input structure

Every structural model is defined in a JSON file. Across the repository, the active input schema uses the blocks UNIT, NODE, ELEM, CONS, MATL, SECT, NDLD, MBLD, SPMV, TPLD, and FABR. This makes the input format readable and easy to edit by hand.

Table A4. Active JSON blocks used by the project

Block	Meaning
UNIT	Defines unit labels for force, length, and temperature
NODE	Stores node coordinates X, Y, and Z
ELEM	Defines element type, material, section, connectivity, and MT release code
CONS	Defines restrained nodal DOFs using binary flags
MATL	Defines E, nu, and temperature coefficient
SECT	Defines A, Iy, Iz, and J
NDLD	Nodal loads F_x, F_y, F_z, M_x, M_y, M_z
MBLD	Member loads in global-vector form
SPMV	Prescribed support movement
TPLD	Temperature load by element
FABR	Fabrication error by element

The current codebase is effectively organized around two active element types: TRUSS and FRAME. In the interfaces layer, these are reduced to the initials T and F. The hybrid solver explicitly supports only these two types, which matches the final bridge model in the notebook.

The CONS block is especially important. In the current implementation, a flag value of 1 means that the corresponding DOF is restrained and therefore eligible to receive a prescribed displacement from SPMV. Because SPMV is filtered through CONS, the user should never prescribe a support movement on a DOF that is not simultaneously restrained.

The ELEM block also uses an MT field for member release handling. For frame elements, the preprocessing stage modifies the local stiffness matrix and equivalent load vectors according to the release condition before assembly.

1.5 Solver architecture and computational workflow

The code follows a clean DSM pipeline. First, the interfaces module reads model data from JSON. Second, element-level quantities are generated, including member length, material and section parameters, transformation matrices, and local stiffness matrices. Third, equivalent nodal loads are generated from member loads, temperature change, and fabrication error. Fourth, the global stiffness matrix and fixed-end force vector are assembled. Fifth, the system is partitioned into free and restrained DOFs, support displacements are imposed, and the reduced linear system is solved. Finally, the postprocess module reconstructs nodal and member results and exports summary tables.

The hybrid solver is the key routine for the final project. It identifies whether each element is type T or F, applies release handling only to frame members, assembles all member contributions into one global system, and then adds fictitious restraints to rotational DOFs with zero stiffness. This prevents singularities that would otherwise arise from truss members assembled in a 12-DOF frame-format system.

1.6 Outputs produced by the code

For each solved case, the postprocessing pipeline produces node-displacement tables, a maximum-displacement summary, support reaction tables, and member-level force/stress summaries. Truss results are exported under the sheet names Node Displacements, Maximum Displacement Summary, Reaction Forces, Member Summary, and Member Full Results. Frame cases use Frame Member Summary and Frame Member Full Results, while mixed truss-frame cases use Hybrid Member Summary and Hybrid Member Full Results. In the hybrid output, truss elements are interpreted as axial members, so their reported local shear-force and end-moment components are zero by definition.

The postprocess module also creates an undeformed model plot and a deformed-shape plot when notebook display is enabled. This makes the project useful both for numerical verification and for quick qualitative checking of model setup.

Table A5. Standard tabular outputs produced by postprocess.py

Output item	Information provided
Node Displacements	Coordinates, translations, rotations, and total displacement magnitude
Maximum Displacement Summary	Location and value of the largest displacement components
Reaction Forces	Computed support reactions at restrained nodes
Member Summary / Frame Member Summary / Hybrid Member Summary	Condensed axial-force and stress summary
Member Full Results / Frame Member Full Results / Hybrid Member Full Results	Expanded element-level local and global end-action output

1.7 Adapting the input for other structural models

To analyze a different structure, the user should first edit the NODE and ELEM blocks so that the geometry and connectivity match the new model. Next, the MATL and SECT blocks should be updated to reflect the desired material and section properties. Then the supports and any imposed support movement should be defined through CONS and SPMV. Finally, NDLD, MBLD, TPLD, and FABR can be edited to create the loading scenario of interest.

For a pure truss model, all elements should be assigned TYPE='TRUSS' and the notebook should call truss_solver. For a pure frame model, all elements should be assigned TYPE='FRAME' and the notebook should call frame_solver. For a mixed model, the user should assign TRUSS and FRAME selectively and use hybrid_solver. The repository already contains examples of all three workflows, so the provided examples can be used as templates for additional models.

2 Technical Validation of the Implementation

The validation portion of the repository includes four truss validation files and six frame validation files, each designed to isolate a specific feature of the implementation. This structure is important because it shows that the code was checked module by module before being applied to the large bridge model. The following examples compare simple analytical expectations with the reported numerical output and therefore provide direct evidence that the major features of the implementation

are working as intended.

2.1 3D truss validation example

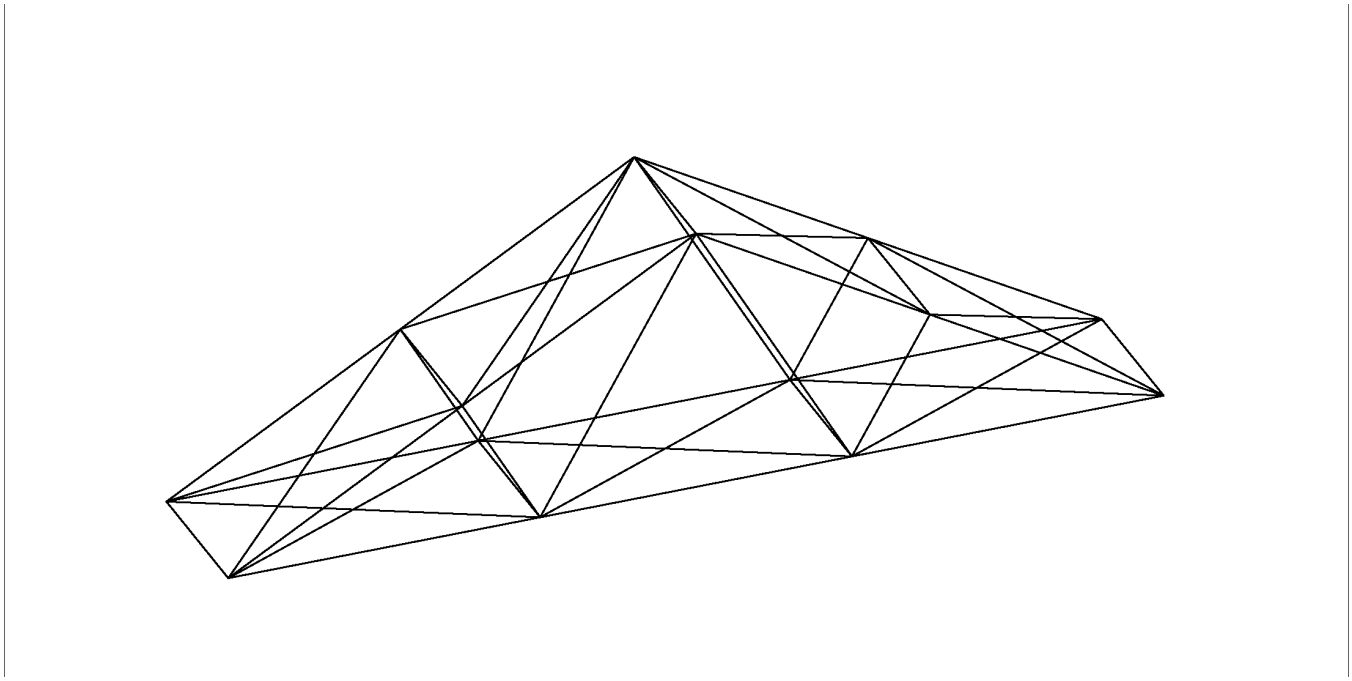

Figure 2: Truss validation model geometry

The truss validation model is a 14-node, 43-member spatial roof-type truss. It is arranged as two longitudinal truss planes connected by transverse ties and bracing members. All members are modeled as truss elements with one material and one section definition. This model is simple enough to describe clearly, but rich enough to demonstrate axial-force transfer in three dimensions.

Table B1. Truss validation cases contained in the repository

Input file	Purpose	Main nonzero input	Selected output
_nodal_loads.json	Nodal loads	Node 11: F_z=-100 kN; node 12: F_x=+100 kN	max $\|u\|$ = 0.0983 m at node 13
_supp_disp.json	Support displacement	Node 1: u_z=+0.01 m	max $\|u\|$ = 0.0134 m at node 7
_temperature.json	Temperature load	All 43 members: TEMP=+30; TEMPCOEF=1×10^{-5}	member axial strain = 3.0×10^{-4}; max $\|u\|$ = 0.00547 m

Input file	Purpose	Main nonzero input	Selected output
_fab_error.json	Fabrication error	Element 6: +0.01 m; element 17: -0.01 m	max $\|u\|$ = 0.0608 m at node 13 and 14

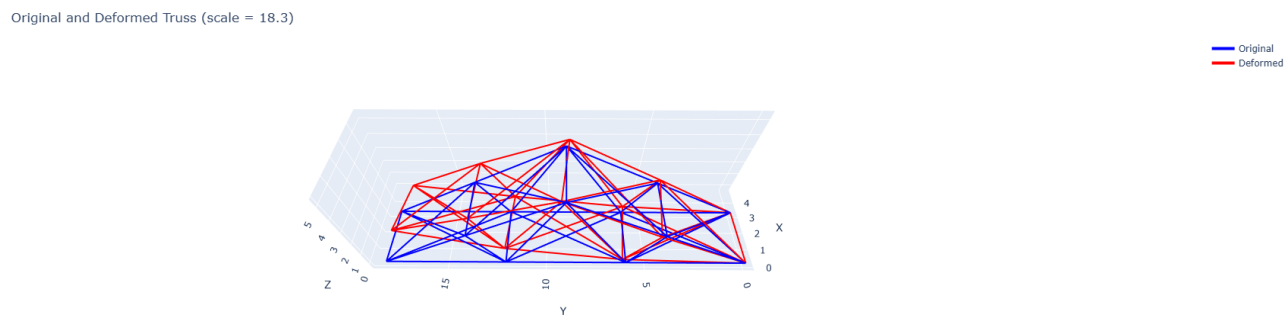

Figure 3: Truss nodal loads

The truss nodal-load case passes a basic but important equilibrium check. The total applied loads are F_x = +100 kN and F_z = -100 kN. The reaction table sums to F_x = -100 kN and F_z = +100 kN, with essentially zero residual in F_y. This confirms that the global force assembly and reaction recovery are consistent.

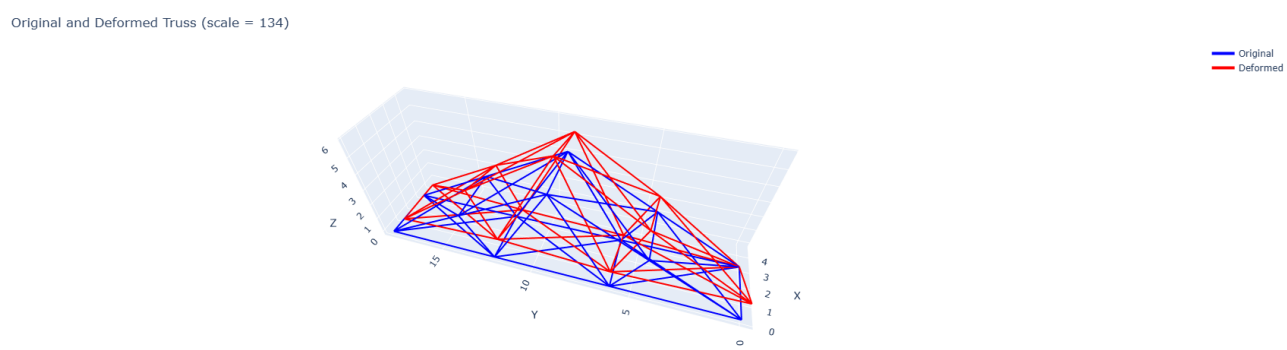

Figure 4: Truss support displacement

The support-displacement case behaves as expected. The prescribed value u_z = +0.01 m at node 1 is reproduced exactly in the node-displacement output, and the largest total displacement in the structure is only 0.0134 m, which is consistent with a model driven by imposed support movement rather than a large external force.

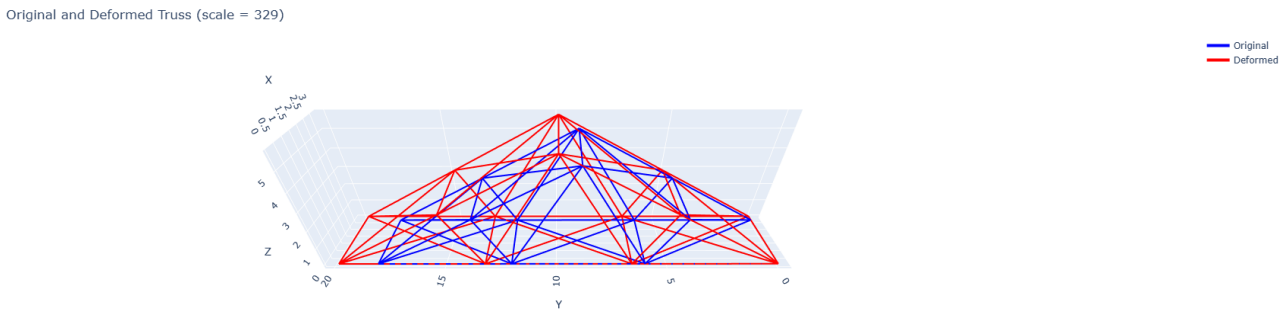

Figure 5: Truss thermal loads

The temperature case is numerically consistent with the implemented load model. Since $\alpha = 1\times10^{-5}$ and $\Delta T = 30$, the theoretical axial strain is 3.0×10^{-4} for every truss member. The exported member summary should therefore report the same value, and for a 6 m member the corresponding free extension is $\Delta L = \alpha \Delta T L = 1.8 \times 10^{-3}$ m. The maximum structural displacement is about 0.00547 m, so this case serves as a compact verification of the temperature-load assembly.

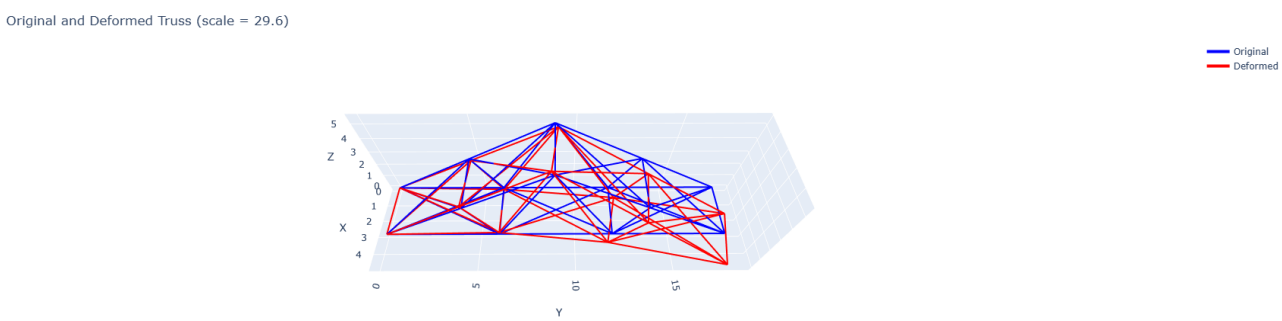

Figure 6: Truss fabrication error loads

The fabrication-error case also serves as an isolated validation case for the fabrication-error implementation. The project assigns +0.01 m to element 6 and -0.01 m to its mirrored counterpart element 17. The output shows a moderate displacement pattern with max $|u| = 0.0608$ m at node 13 and 14, consistent with the idea that local length incompatibilities are being transformed into equivalent element actions and redistributed through the truss.

Table B2. Representative truss validation checks

Check	Hand expectation	Observed code behavior	Assessment
Nodal-load equilibrium	+100 kN in x; -100 kN in z	Reactions sum to -100 kN in x and +100 kN in z	Pass

Check	Hand expectation	Observed code behavior	Assessment
Prescribed support motion	Node 1: $u_z = +0.01$ m	Node-displacement table reports $u_z = +0.01$ m at node 1	Pass
Thermal strain relation	axial strain = $1 \times 10^{-5} \times 30 = 3.0 \times 10^{-4}$	Member summary reports axial strain = 3.0×10^{-4}	Pass
Fabrication-error strain check	elements 6 and 17 should give axial strain $\pm 0.01/6 = \pm 1.667 \times 10^{-3}$	Member summary reports axial strain $\approx +1.6667 \times 10^{-3}$ for element 6 and $\approx -1.6667 \times 10^{-3}$ for element 17	Pass

2.2 3D frame validation example

The frame validation model is a 20-node, 32-element, four-column space frame with an 8 m by 8 m plan and five elevation levels from z = 0 m to z = 24 m. Nodes 1 to 4 are fully restrained at the base, while the upper levels are connected by beams in each floor plane. This is a good validation model because it contains true frame behavior, allows member-end release checks, and is easy to embed in the 3D formulation even when the hand solution is simplified.

Table B3. Frame validation cases contained in the repository

Input file	Purpose	Main nonzero input	Selected output
_nodal_loads.json	Nodal moments	Selected M_x and M_y nodal moments	max $\|u\| = 1.138$ m at node 20
_member_loads.json	Distributed member load	12 members: qx = +25 kN/m	max $\|u\| = 12.764$ m at node 20
_moment_release.json	Release verification	Same qx with selected MT = 1 or 2	max $\|u\| = 16.353$ m at node 19
_supp_disp.json	Support displacement	Node 1 and node 3 support movements	max $\|u\| = 0.0366$ m at node 19
_temperature.json	Temperature load	All 32 members: TEMP = -30; TEMPCOEF = 1×10^{-5}	member axial strain = -3.0×10^{-4}
_fab_error.json	Fabrication error	Element 5 and 6: ERROR = -0.01 m	max $\|u\| = 0.0211$ m at node 17

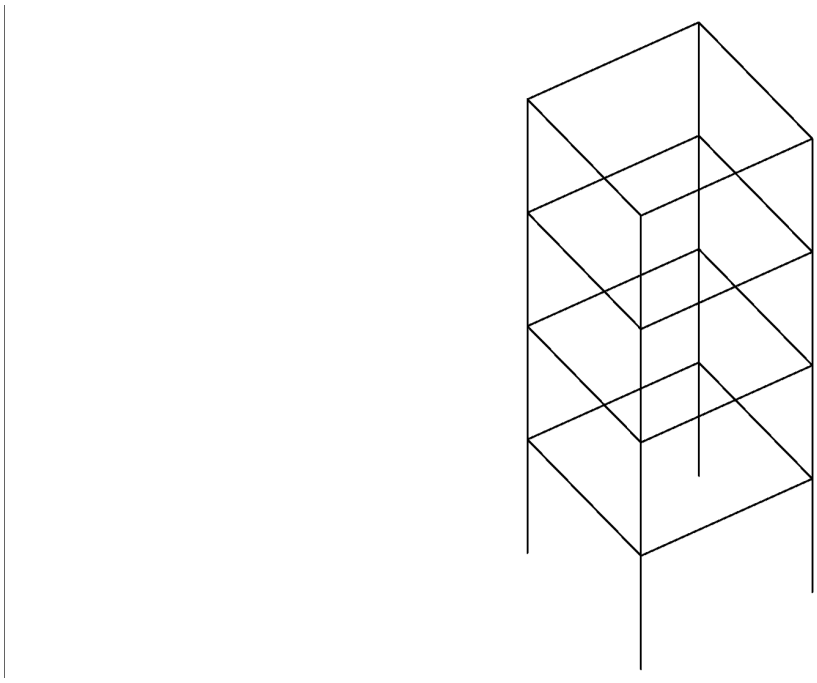

Figure 7: Frame validation model geometry

The frame nodal load case illustrates the deformation behavior in different directions when nodal moments are applied respectively in different directions at the two intermediate floors, corresponding respectively to deformation in the y direction and deformation in the x direction.

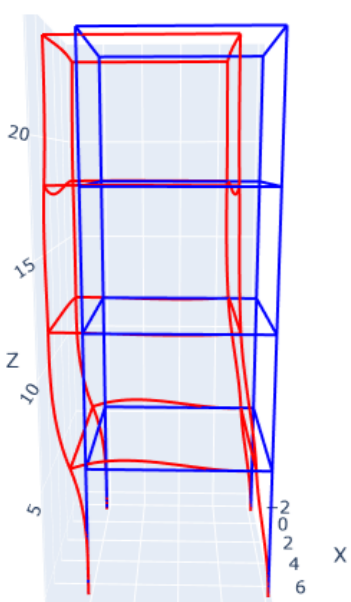

Figure 8: Nodal Loads on Second Layer y axis

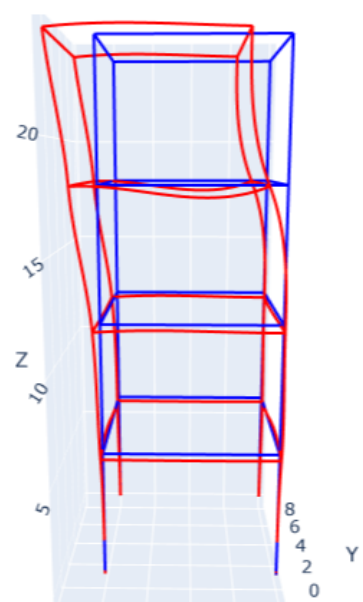

Figure 9: Nodal Loads on third Layer x axis

The frame member-load case gives a strong load-balance check. The selected member loads produce a total x-direction resultant of 2000 kN, and the reaction table sums to -2000 kN in F_x. This confirms

that the member-load conversion and reaction recovery are consistent with the specified input.

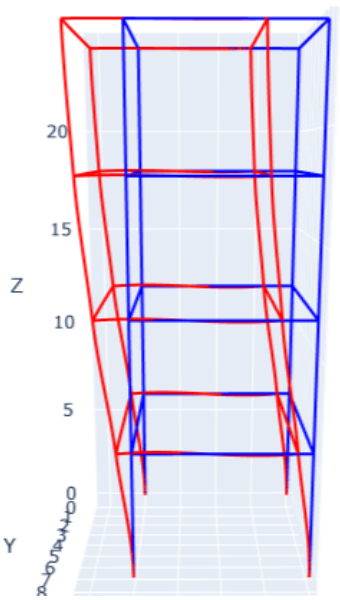

Figure 10: Pure Member Loads

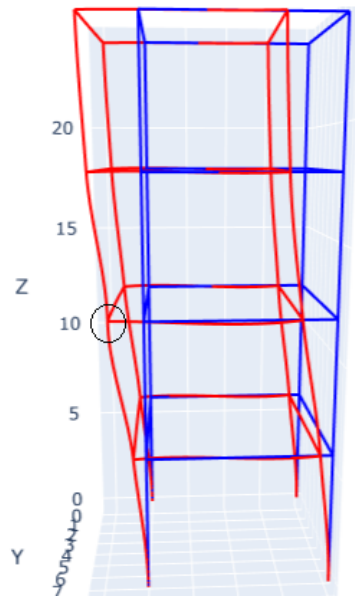

Figure 11: Moment Released

The moment-release case shows the expected reduction in stiffness. The maximum total displacement increases from 12.764 m in the unreleased member-load case to 16.353 m in the released case, which is an increase of about 28.1 percent. This is exactly the trend that should occur when selected member ends are released.

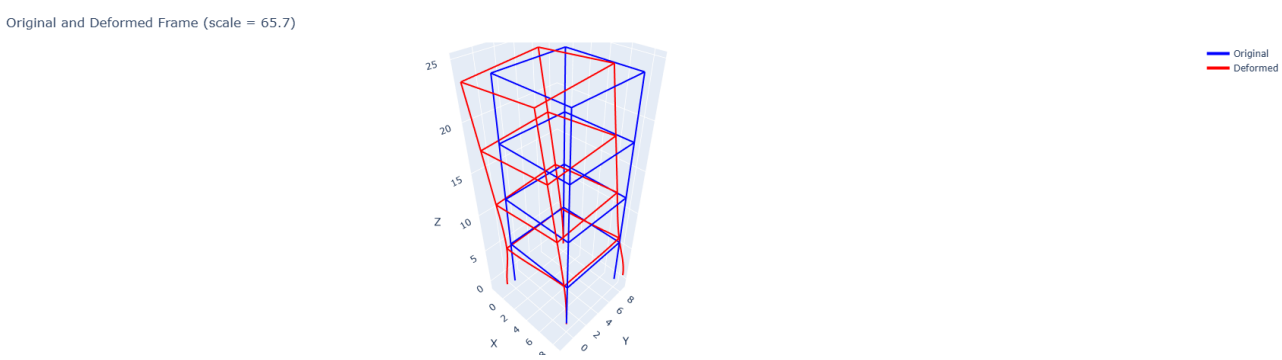

Figure 12: Frame support displacement

The support-displacement case again satisfies the imposed-motion check: node 1 is prescribed to move by (-0.01, -0.01, -0.01) m and node 3 by (+0.01, +0.01, +0.01) m, and the node-displacement table reflects those values at the restrained nodes.

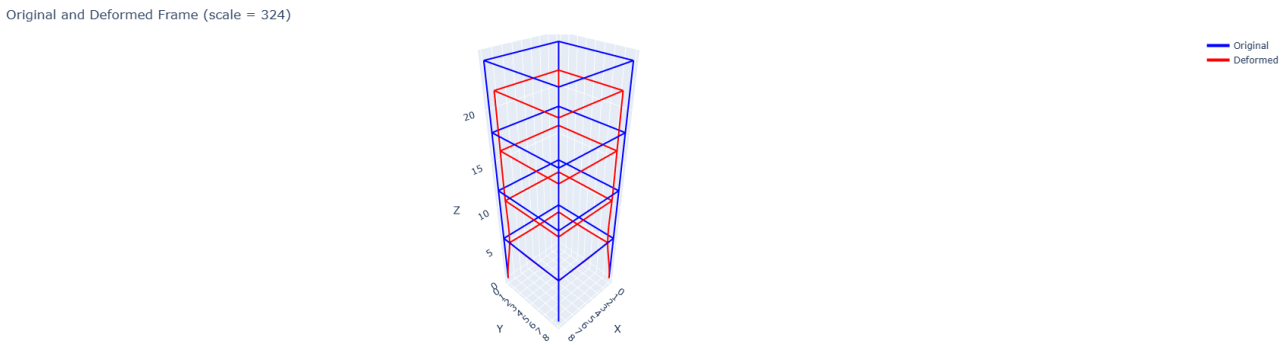

Figure 13: Frame temperature change

The temperature file uses $\Delta T = -30$ with $\alpha = 1 \times 10^{-5}$, so the theoretical axial strain is -3.0×10^{-4}. The frame member results should therefore show the same value. This case provides a direct verification of the frame temperature-load implementation.

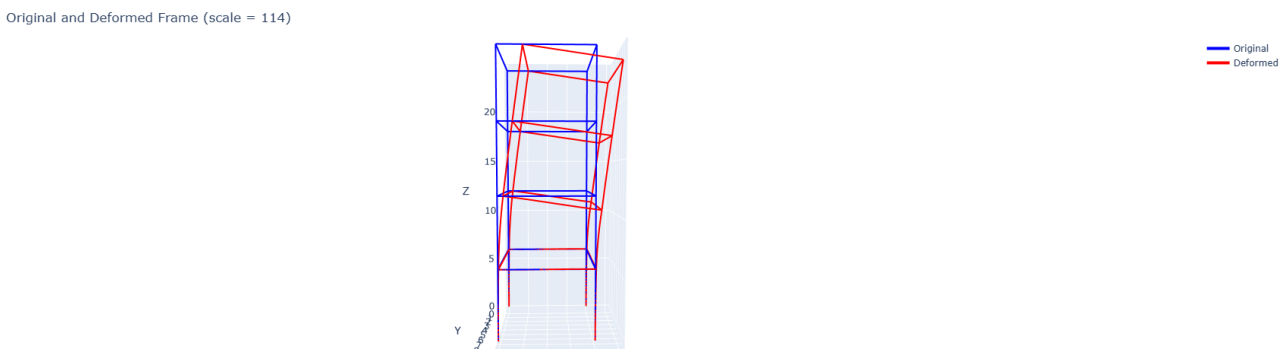

Figure 14: Frame fabrication error loads

The fabrication-error case also provides a focused validation of the fabrication-error formulation in the frame model. The project assigns -0.01 m to both element 5 and 6. The output shows a moderate displacement pattern with max $|u| = 0.0211$ m at node 17, consistent with the idea that local length incompatibilities are being transformed into equivalent element actions and redistributed through the frame.

Table B4. Representative frame validation checks

Check	Hand expectation	Observed code behavior	Assessment
Distributed-load balance	Total applied x-load = 2000 kN	Reactions sum to -2000 kN in F_x	Pass

Check	Hand expectation	Observed code behavior	Assessment
Pure nodal-moment equilibrium	No translational load resultant; only balancing support moments should appear	Reactions sum to approximately 0 in F_x, F_y, and F_z, while support moments balance the imposed nodal moments	Pass
Release flexibility trend	Released case should be softer	max $\lvert u \rvert$ rises from 12.764 m to 16.353 m	Pass
Prescribed support motion	Node 1 and 3 support values should be exact	Reported restrained-node displacements match input	Pass
Thermal strain relation	axial strain = $1 \times 10^{-5} \times$ (-30) = -3.0×10^{-4}	Member summary reports axial strain = -3.0×10^{-4}	Pass
Fabrication-error compatibility	For element 5 or 6, expected axial strain = $-0.01/6 = -1.667 \times 10^{-3}$	Member summary reports axial strain $\approx -1.666489 \times 10^{-3}$ for elements 5 and 6	Pass

Table B4 therefore covers all six main frame validation features used in this project: nodal loads, member loads, end release, prescribed support displacement, temperature change, and fabrication error.

2.3 Overall validation assessment

Taken together, the truss and frame validation files provide a solid implementation check. The repository does not rely on one large benchmark alone; instead, it isolates the major features of the solver in separate cases, which makes the numerical behavior of each module easier to verify and interpret.

3 Modeling Study of a Complex 3D Structure

The final part of the report presents the analysis of a large mixed-element bridge structure. In the notebook, this is presented as the final complex-structure study and solved with the hybrid solver. The model is more substantial than the earlier validation examples and better demonstrates the intended purpose of the code: the analysis of systems containing both truss and frame members.

3.1 Background and structural choice

The selected case study is a three-span continuous arch-truss bridge consistent with the Jiujiang Yangtze River Bridge configuration used in this study. The notebook describes the model as a three-dimensional three-span continuous arch-truss bridge composed of two longitudinal main truss planes, transverse connecting members, web bracing, and arch ribs arranged across all three spans.

The key geometric information embedded in the project is as follows: the total bridge length is 576 m, the span arrangement is 180 + 216 + 180 m, the transverse spacing between the two main truss planes is 12.5 m, and the longitudinal panel length is 9 m. In the arch-rib idealization, the side spans use a 90 m arch rib with 24 m rise, while the main span uses a 126 m arch rib with 32 m rise. This gives rise-to-span ratios of about 1/3.75 and 1/3.94, respectively.

This bridge is suitable for the present study because it naturally motivates a mixed-element model. Most of the bracing system can still be idealized as axial-force members. However, the arch ribs and the longitudinal members located at the $z = 8$ m and $z = 16$ m levels are treated as frame members so that their bending stiffness and end moments can be represented.

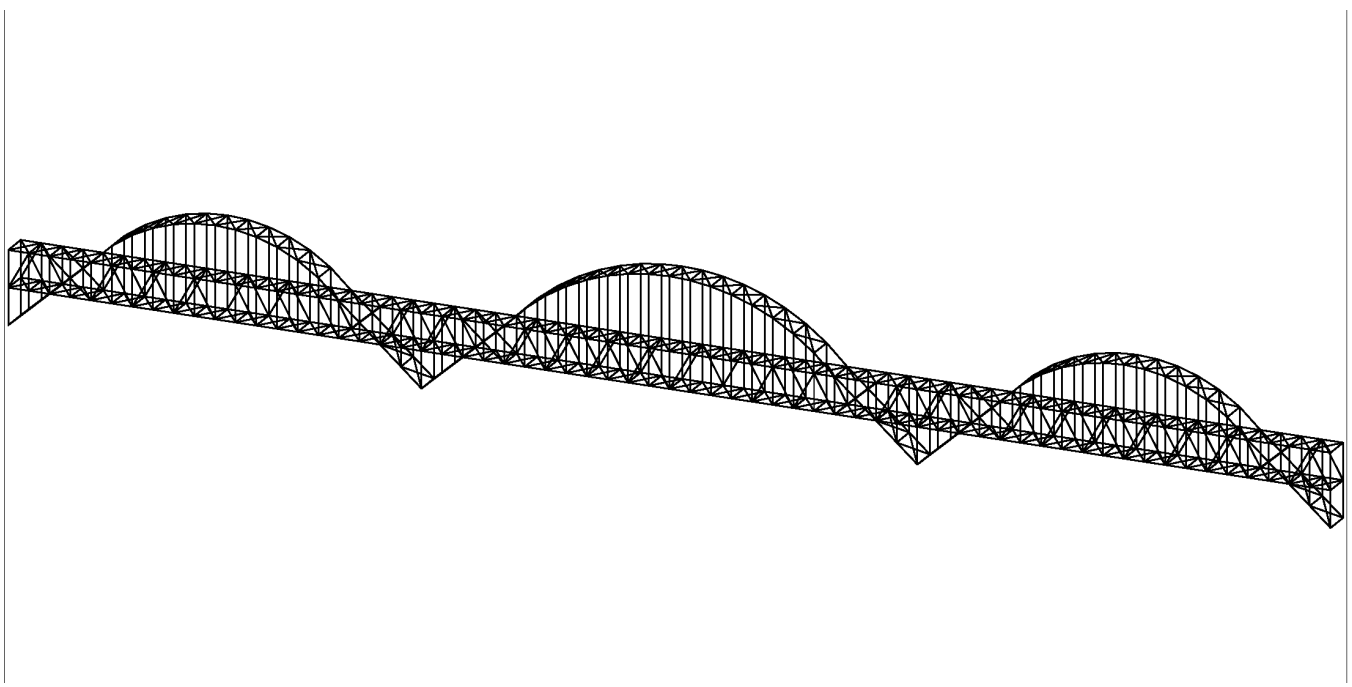

Figure 15: DXF-based idealized bridge geometr

3.2 Analytical idealization in the code

The final bridge JSON files define a model with 354 nodes and 1225 elements. Of these, 1005 are truss elements and 220 are frame elements. The 220 frame elements include the arch ribs and the selected longitudinal members at the $z = 8$ m and $z = 16$ m levels. The model therefore uses frame

elements selectively rather than everywhere, which is consistent with the intended mixed-element idealization.

Table C1. Size and composition of the final bridge model

Model property	Value
Nodes	354
Elements	1225
Truss elements	1005
Frame elements	220
Support nodes listed in CONS	8
Main global x-range	0 m to 576 m
Main global y-range	0 m to -12.5 m
Main global z-range	0 m to 64 m

Table C2. Support lines used in the final bridge model

Support nodes	Location	Interpretation
1 and 4	x = 0 m	Uz restrained; end support line
109 and 112	x = 180 m	Selective Ux/Uy/Uz restraint at internal support line
241 and 244	x = 396 m	Uz restrained; internal support line
349 and 352	x = 576 m	Uz restrained; end support line

3.3 Final-structure load cases considered in this study

The analysis does not begin with a single combined load case. Instead, it first studies the final bridge under separate nodal-load, member-load, support-displacement, and temperature-load files before running the comprehensive combined case. This separation helps distinguish the response contributions of different loading types.

Table C3. Final bridge cases analyzed in this study

Input file	Role	Main loading description	Selected output
_nodal_loads.json	Nodal loads only	130 loaded nodes with concentrated downward loads	max $\lvert u \rvert = 0.0538$ m at node 37
_member_loads.json	Member loads only	350 members with q_global = [0, -1, 0] kN/m	max $\lvert u \rvert = 49.209$ m at node 349
_supp_disp.json	Support displacement only	4 internal support nodes with $u_z = +0.01$ m	max $\lvert u \rvert = 0.0114$ m at node 202
_temperature.json	Temperature only	All members: TEMP = +50 and TEMPCOEF = 1×10^{-5}	max $\lvert u \rvert \approx 0.19874$ m at node 354
_comprehensive_loads.json	Combined case	50 nodal loads, 40 loaded members, 8 support movements, and a uniform +50°C temperature rise on all members	max $\lvert u \rvert = 3.2791$ m at node 350; response still dominated by mechanical loading

3.4 Detailed discussion of the comprehensive bridge case

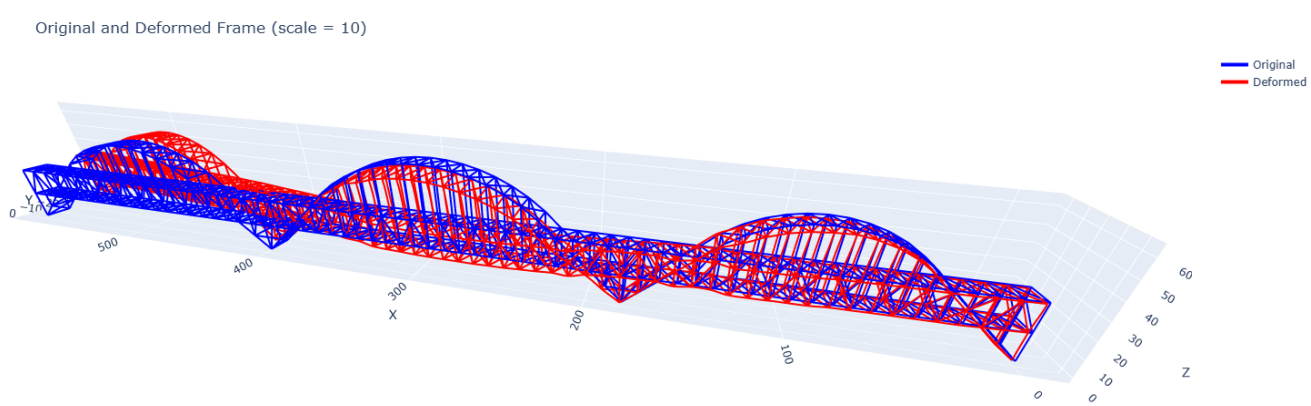

Figure 16: Comprehensive-case deformed shape

The comprehensive file is the strongest basis for the final discussion because it combines multiple realistic actions in a single run. In this model, this file includes a uniform +50°C temperature rise on all members. The JSON file contains 50 nonzero nodal loads with a total vertical load of -3525 kN. It also contains 40 members with a vertical distributed load q_global = [0, 0, -120] kN/m. Since the loaded members are 9 m long, the total vertical resultant of these member loads is -43,200 kN. The total vertical applied mechanical load is therefore -46,725 kN, and the reaction table sums to +46,725 kN in F_z. This provides an excellent global equilibrium check for the mixed solver. The

uniform temperature term produces self-equilibrated element actions but does not alter this net vertical load balance.

Table C4. Global vertical equilibrium check for the comprehensive bridge case

Quantity	Value
Total nodal vertical load	-3525 kN
Total distributed-member vertical load	-43200 kN
Total applied vertical load	-46725 kN
Sum of support reactions in F_z	+46725 kN

The detailed node- and member-ranking tables below are taken from the Excel output for the comprehensive case with a uniform +50°C temperature rise applied to all members. They therefore reflect the model configuration with $\alpha = 1 \times 10^{-5}$.

Table C5. Nodes with the largest total displacement in the comprehensive case

Node	x (m)	y (m)	z (m)	u_x (m)	u_y (m)	u_z (m)	$\|u\|$ (m)
350	576.0000	0.0000	16.0000	0.6537	-3.2130	0.0443	3.2791
353	576.0000	-12.5000	16.0000	0.4843	-3.2192	-0.0113	3.2554
351	576.0000	0.0000	32.0000	0.6370	-3.1868	0.0521	3.2502
349	576.0000	0.0000	0.0000	0.6791	-3.1753	0.0400	3.2473
354	576.0000	-12.5000	32.0000	0.4834	-3.1930	-0.0031	3.2294
352	576.0000	-12.5000	0.0000	0.4841	-3.1815	-0.0200	3.2182

The largest displacements occur near the right end of the bridge around x = 576 m, especially at nodes 349 to 354. In the comprehensive case, the maximum total displacement is 3.2791 m at node 350, and it is dominated by the negative y-direction component $u_y = -3.2130$ m together with a smaller positive x-direction drift $u_x = 0.6537$ m. This is physically plausible because the right end is directly affected by both distributed loading and nonuniform support movement, while the thermal term is modest because $\alpha = 1 \times 10^{-5}$.

The largest vertical displacement occurs at node 181, where $u_z = -0.7352$ m at coordinates $(297, 0, 16)$. This is near the central part of the bridge rather than directly at a support. That pattern is physically reasonable: under the combined load case, the bridge sags most strongly in the more flexible span region, whereas the largest total displacement near the right end is governed mainly by lateral drift and support-kinematic effects.

The hybrid member output distinguishes the physical interpretation of truss and frame elements. Truss elements retain their axial force N, but their local shear forces and end moments are reported as zero because those quantities are not physical response quantities for truss members. Frame elements retain the full local end-force and end-moment output.

Table C6. Members with the largest axial force magnitude in the comprehensive case

Element	Type	i	j	L (m)	Axial force N (kN)	Max end moment (kN m)
826	T	237	243	9.0000	9238.4262	0.0000
847	T	243	249	9.0000	8991.7389	0.0000
387	T	111	117	9.0000	8786.0366	0.0000
366	T	105	111	9.0000	8536.4116	0.0000
761	F	219	224	12.5568	-8364.0405	337.8158
774	T	223	232	24.0832	7591.5595	0.0000

For the type T rows in Table C6, the zero moment values indicate an axial-only truss interpretation rather than a bending-demand ranking. Bending demand is therefore evaluated separately using only frame elements in Table C7.

Table C7. Frame members with the largest end moments in the comprehensive case

Frame element	i	j	L (m)	Axial force (kN)	Max end moment (kN m)
421	121	125	9.0000	3111.2671	2221.2834
788	227	231	9.0000	3623.3702	2204.1966
328	95	99	9.0000	-1002.0915	2128.9521
881	253	257	9.0000	-1367.5582	2111.9250
692	199	205	9.0000	6071.3961	1885.1521

The largest axial-force magnitudes are concentrated in a few chord and diagonal members near the major force-transfer zones. The single largest value is 9238.4 kN in truss element 826 between nodes 237 and 243, located from $(387, 0, 32)$ to $(396, 0, 32)$ immediately adjacent to the internal support line at $x = 396$ m. Nearby members 847, 818, and 800 on the same right-side region also rank highly, while members 366 and 387 form another strong cluster around $x = 171$ m to $x = 189$ m. For the truss rows in this ranking, the end-moment column is zero because those members are axial-only

members. The ranking confirms strong axial-force redistribution through the truss-and-arch load path under the combined mechanical and thermal actions.

In the frame-member set, the largest frame end moments are concentrated in the longitudinal frame members at the $z = 16$ m level. The largest frame end moment in the comprehensive case is 2221.3 kN m in element 421, which lies between nodes 121 and 125 from $(198, 0, 16)$ to $(207, 0, 16)$. The next entries, elements 788, 328, 881, and 692, are also 9 m longitudinal members at $z = 16$ m. This indicates that the mixed frame idealization captures bending demand in the longitudinal chord system as well as in the arch ribs.

3.5 Interpretation of structural behavior

Several broad behavioral patterns are visible in the comprehensive output.

First, the largest axial forces cluster in support-adjacent chord and diagonal members, especially around $x \approx 396$ m, with a second strong cluster around $x \approx 180$ m. The ranking includes both strong tension and strong compression, which is consistent with the way an arch-truss bridge redistributes combined load through coupled chord, diagonal, and arch action.

Second, the largest frame end moments are concentrated in the longitudinal frame members at the $z = 16$ m level. This is consistent with the mixed-element idealization because these longitudinal members are represented with bending stiffness and therefore carry end moments in addition to axial force.

Third, the maximum global displacement does not occur in the most massive arch-rib region, but instead near the right end support line at node 350, where the combined support movement and distributed loading induce a larger lateral drift. This is physically meaningful because the response of a continuous bridge is influenced not only by span gravity action but also by support kinematics and continuity effects.

Fourth, the comparison among the separate final-structure cases is informative. The nodal-load-only case produces a very small maximum displacement of only 0.0538 m, while the support-displacement-only case produces 0.0114 m. The member-load-only case is much more severe, indicating that the distributed loading pattern used in that file dominates the lateral response. The dedicated temperature-only case produces about 0.19874 m, corresponding to the uniform +50°C temperature rise with $\alpha = 1 \times 10^{-5}$. The comprehensive case then combines vertical member load, nodal load, support movement, and the same uniform +50°C temperature rise, and its maximum total displacement is 3.2791 m at node 350. With the thermal action included, the overall behavior is still governed primarily by the mechanical loading pattern.

These observations show that the bridge model is useful for more than merely printing output tables. The bridge model is being used to interpret load path, identify critical regions, and justify the mixed idealization of the structure.

3.6 Modeling limitations

Only one material and one section set are used throughout the bridge model. Mixed-section models have been tested. While they run successfully, the deformation results are not typical.

A uniform linear thermal-expansion coefficient of 1×10^{-5} is used throughout the model. The applied temperature field is still highly simplified because it is uniform over all members.

The bridge deck, floor system, wind bracing details, and connection flexibility are simplified considerably.

The global stiffness matrix is assembled and stored in dense form. This is appropriate for the current validation models and the present bridge study, but larger practical bridge analyses would benefit from sparse or banded storage together with bandwidth-reducing node-numbering strategies to improve both memory efficiency and solution speed.

The comprehensive case is a strong mixed-loading demonstration, but it is still not a code-calibrated design-load combination.

Conclusion

This project provides a coherent Direct Stiffness Method framework for three-dimensional structural analysis using truss, frame, and mixed truss-frame idealizations. The implementation is well organized, and the computational workflow is straightforward, and the JSON input system makes it straightforward to define and modify structural models.

The validation studies show that the implementation handles the major features it claims to support, including nodal loads, member loads, support displacements, temperature loads, fabrication errors, and frame moment release. The temperature cases are also consistent with the intended verification scale of the model. The final bridge study then demonstrates that the hybrid solver can be used on a much larger mixed system with both axial-only members and bending-active frame members, and that the results can be interpreted in a structurally meaningful way. At the same time, the report indicates clear directions for further development on larger practical models, including sparse-storage techniques, improved node numbering, and an explicit statement of the frame local-axis convention.

Taken together, the user guide, validation studies, and final bridge application show that the project provides a complete and coherent workflow for three-dimensional truss, frame, and hybrid structural analysis within the scope of the current implementation.

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